RYAN M. KIM

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EDUCATION

Northeastern University Expected 2028
B.S. in Electrical and Computer Engineering, Dean's List *Boston, MA*

- **Coursework:** Digital Design and Computer Organization, Embedded Design: Enabling Robotics, Discrete Structures, Circuits and Signals, Fundamentals of Algorithms, Fundamentals of Computer Science I, Physics I/II, Calculus III, Differential Equations and Linear Algebra

TECHNICAL SKILLS

Programming: Python, C/C++, Racket, Bash, MATLAB, NumPy, Pandas, PyTorch
Hardware: Xilinx Vivado, Quartus Prime, SystemVerilog, Verilog, RISC-V Assembly, LTspice
Tools: Git, Linux, RARS, LaTeX, Markdown, Oscilloscope, Multimeter

RESEARCH EXPERIENCE

Northeastern University Computer Architecture Research Jan. 2026 – Present
Undergraduate Researcher *Boston, MA*

- Optimized matrix multiplications by distributing workloads to gain experience implementing parallel computing techniques in C/C++.
- Developed parallelized multi-threaded matrix multiplication in OpenMP. Using POSIX threads, learned to managed concurrent memory access and workloads across multiple threads.
- Implemented Strassen's algorithm using OpenMP, utilizing recursive task parallelism to reduce matrix multiplication complexity.

UMass Chan Medical School, Dept. of Biochemistry and Molecular Biotechnology Aug. 2022 – Sep. 2024
Research Assistant, Schiffer Lab *Worcester, MA*

- Investigated structural basis of influenza virus entry using cryo-electron tomography and deep learning.
- Trained convolutional neural networks to automatically segment viral protein components, processing 100+ tomograms and enabling subnanometer structural determination of 12,000+ particles at 8–10 Å resolution.
- Contributed to quantitative morphology analysis pipeline using scikit-learn, Open3D, and EMAN2.99
- Developed and debugged data pipelines in Python using Numpy, Pandas, and Matplotlib.

PUBLICATION

Q.J. Huang, R. Kim, K. Song, N. Grigorieff, J.B. Munro, C.A. Schiffer, & M. Somasundaran. "Virion-associated influenza hemagglutinin clusters upon sialic acid binding visualized by cryoelectron tomography." *Proc. Natl. Acad. Sci. U.S.A.* 122(16): e2426427122 (2025). doi.org/10.1073/pnas.2426427122

PROJECTS

RISC-V CPU | *SystemVerilog, Xilinx Vivado, FPGA* 2026

- Implemented a RISC-V-like single-cycle CPU in SystemVerilog targeting a Xilinx PYNQ-Z2 FPGA.
- Developed key datapath and control components, including an ALU, register file, instruction decoder, instruction memory, data memory, and program counter logic.
- Extended the processor with branching support to execute nontrivial assembly programs on hardware.
- Validated the design through Xilinx Vivado simulation and testing using hardware debugging interfaces.

Parallel Image Filtering Tool | *C++, OpenMP* 2026

- Developed a OpenMP program that partitions images into tiles and processes tiles concurrently to accelerate extraction, filtering, and file output.
- Designed and implemented image filters that are used in parallel, such as grayscale, threshold, sharpen, edge, and Gaussian blur operations.
- Used thread-level parallelism and local memory buffers to avoid race conditions during tile processing and writing.

RNA Translator | *C++, Python, OpenMP* 2026

- Translated RNA sequences to a chain of amino acid using C++, OpenMP and parallel programming concepts, validated on the SARS-CoV-2 genome.

ENGAGEMENT EXPERIENCE

Youth For Community Improvement 2021 – 2023
Grant Committee Member, Greater Worcester Foundation *Worcester, MA*

- Evaluated grant applications and allocated $40,000 in funding to nonprofit organizations addressing community issues through a youth-led program, facilitating meetings with community speakers and authoring advocacy briefs to inform grant decision-making